Isotopes	^{20}Ne	^{40}Ar	^{84}Kr	^{132}Xe	^{211}Rn, ^{220}Rn, ^{222}Rn
Atomic Mass ($g.mol^{-1}$ at 293 K)	20.18	39.95	77.92	131.29	[222]
Electronic configuration	$[He]2s^2 2p^6$	$[Ne]3s^2 3p^6$	$[Ar]3d^{10} 4s^2 4p^6$	$[Kr]4d^{10} 5s^2 5p^6$	$[Xe]\ 4f^{14}\ 5d^{10} 6s^2\ 6p^6$
Atomic radius (Å)	1.54	1.88	2.02	2.16	2.20
Density ($g.cm^{-3}$ at 293 K)	8.25×10^{-4}	1.63×10^{-3}	3.42×10^{-3}	5.37×10^{-3}	9.07×10^{-3}
Melting point (K)	24.56	83.81	115.78	161.4	202
Boiling point (K)	27.104	87.30	119.74	165.05	211.5

Table 3.10 Physical properties of group 18 elements

Properties of inert gases:

Physical properties:

Noble gases are monoatomic, odourless, colourless, tasteless, and non-inflammable. They are highly unreactive. They are non-metallic in nature.

Chemical Properties:

Only the xenon and krypton show some chemical reactivity. Xenon fluorides are prepare by direct reaction of xenon and fluorine under different conditions as shown below.

$$Xe + F_2 \xrightarrow[400^0C]{Ni} XeF_2$$

$$Xe + 2F_2 \xrightarrow[400^0C]{Ni\ /acetone} XeF_4$$

$$Xe + 3F_2 \xrightarrow[400^0C]{Ni\ /200\ atm} XeF_6$$

When XeF_6 is heated at 50 °C in a sealed quartz vessel it forms $XeOF_4$.

$$2XeF_6 + SiO_2 \xrightarrow{50^0C} 2XeOF_4 + SiF_4$$

When the reaction is continued the following reaction takes place.

$$2XeOF_4 + SiO_2 \longrightarrow 2XeO_2F_2 + SiF_4$$

$$2XeO_2F_2 + SiO_2 \longrightarrow 2XeO_3 + SiF_4$$

On hydrolysis with water vapour XeF_6 gives XeO_3

$$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$$

When XeF_6 reacts with 2.5 M NaOH, sodium per xenate is obtained.

$$2XeF_6 + 16NaOH \longrightarrow Na_4XeO_6 + Xe + O_2 + 12NaF + 8H_2O$$

Sodium per xenate is very much known for its strong oxidizing property. For example, it oxidises manganese (II) ion into permanganate ion even in the absence of the catalyst.

$$5XeO_6^{4-} + 2Mn^{2+} + 14H^+ \longrightarrow 2MnO_4^- + 5XeO_3 + 7H_2O$$

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